the simplified text), and 91.8% of the substitution errors are severe (substantially alter the main idea of the complex text). Out of 49 samples which have verifiable hallucinated terms, 65.3% of hallucinated terms occur in the pretraining data.

6 Discussion and Future Work

We briefly discuss our findings, and offer some guiding principles for future work on building more factual large language models.

Downstream impact of model hallucinations. LLMs are now used in several user-facing applications, and past work has highlighted the downstream harms made possible by model hallucinations, including in AI-powered search tools (Raji et al., 2022), and in code generation (Lanyado, 2023; Claburn, 2024). Our benchmark aims to provide a comprehensive and rigorous measurement of the extent to which LLMs hallucinate, to enable progress on building more trustworthy models.

What will it take to have truthful AI systems? This work shows that LLM hallucinations may arise from multiple sources in the training data—ranging from incorrect information in the pretraining data, to total fabrication in model generations. Since models hallucinations do not seem to have a single isolated cause, we speculate that effective hallucination mitigation would require multiple complementary approaches. For example, a retrieval-based backbone could be effective for long-tailed information, but not when the datastore does not have relevant information to begin with. Approaches which require LLMs to verbalize uncertainty may be more effective in such scenarios. However, while these are likely to patch a portion of hallucination errors, our findings also indicate that current LLMs make semantic errors even when the context is completely provided as in the case of summarization, indicating the need for more robust frameworks for semantic meaning overall.

Causal attributions. In this work, we take a step towards tracing back hallucinations to training data. Future work would construct causal frameworks, to study counterfactual questions about the inclusion of specific datapoints and their effect on specific model hallucinations to shed more light on the root cause of hallucination. In addition, while we search for facts as they are stated in model responses, these facts could be present implicitly in pretraining corpora. Future work would attribute hallucinations by computing these implicit inferences as well.

7 Conclusion

In this work, we study hallucination in generative large language models. We contribute a high-quality resource, **HALoGEN**🔦, to measure and identify model hallucinations in a broad range of scenarios. Using **HALoGEN**🔦, we are then able to create a large-scale dataset of hallucinations from 150,000 large-language model generations, sourced from 14 different language models. We use this dataset to systematically trace back language model hallucinations to their training data, and proposing a classification schema for three types of hallucination errors. Our work highlights how nuanced the causes of LLM hallucination can be, and we discuss potential strategies to mitigate hallucination in large-language models based on the type of errors models make. We hope our framework provides the foundation for scientific study of hallucination in large language models.

8 Limitations

HALoGEN🔦 aims to provide a broad-coverage hallucination benchmark for a range of NLP use cases. While the automated hallucination detection approaches used in this work enable scalable evaluation, the reliability of our benchmark scores are limited by the accuracy of these underlying techniques. For use cases like code generation, our automated verifiers are more accurate since they perform an exact search against a library of available Python packages; on the other hand, open-ended generation tasks are more subjective and challenging to evaluate. As automated hallucination evaluations improve, these techniques can be incorporated into **HALoGEN**🔦.

An additional limitation relates to training data attribution. While WIMBD enables search over widely-used open-source pretraining corpora, many of the LLMs examined in this work do not release their data sources, limiting the accuracy of our attributions. This points toward the need for open language models (Groeneveld et al., 2024; Mehta et al.; Biderman et al., 2023) which enable transparent inspection of pretraining data.

Finally, while our work provides a framework to measure both factual precision and appropriate model abstention, our metrics do not account for coverage— whether the model response contains all the information it should. Future work would introduce methodologies to measure coverage, as well as further improve the accuracy of verifiers.

Acknowledgment

The authors would like to thank Aakanksha Naik and Ronan Le Bras for helpful discussions regarding this work. The 'historical events' category in **HALoGEN** was inspired by an example of a model hallucination in a 2023 New York Times article (Weise and Metz, 2023). This research was supported by the NSF DMS-2134012, ONR N00014-24-1-2207, and the Allen Institute for AI.

References

OpenAI Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, Red Avila, Igor Babuschkin, Suchir Balaji, Valerie Balcom, Paul Baltescu, Haiming Bao, Mo Bavarian, Jeff Belgum, Irwan Bello, Jake Berdine, Gabriel Bernadett-Shapiro, Christopher Berner, Lenny Bogdonoff, Oleg Boiko, Madelaine Boyd, Anna-Luisa Brakman, Greg Brockman, Tim Brooks, Miles Brundage, Kevin Button, Trevor Cai, Rosie Campbell, Andrew Cann, Brittany Carey, Chelsea Carlson, Rory Carmichael, Brooke Chan, Che Chang, Fotis Chantzis, Derek Chen, Sully Chen, Ruby Chen, Jason Chen, Mark Chen, Benjamin Chess, Chester Cho, Casey Chu, Hyung Won Chung, Dave Cummings, Jeremiah Currier, Yunxing Dai, Cory Decareaux, Thomas Degry, Noah Deutsch, Damien Deville, Arka Dhar, David Dohan, Steve Dowling, Sheila Dunning, Adrien Ecoffet, Atty Eleti, Tyna Eloundou, David Farhi, Liam Fedus, Niko Felix, Sim'on Posada Fishman, Juston Forte, Isabella Fulford, Leo Gao, Elie Georges, Christian Gibson, Vik Goel, Tarun Gogineni, Gabriel Goh, Raphael Gontijo-Lopes, Jonathan Gordon, Morgan Grafstein, Scott Gray, Ryan Greene, Joshua Gross, Shixiang Shane Gu, Yufei Guo, Chris Hallacy, Jesse Han, Jeff Harris, Yuchen He, Mike Heaton, Johannes Heidecke, Chris Hesse, Alan Hickey, Wade Hickey, Peter Hoeschele, Brandon Houghton, Kenny Hsu, Shengli Hu, Xin Hu, Joost Huizinga, Shantanu Jain, Shawn Jain, Joanne Jang, Angela Jiang, Roger Jiang, Haozhun Jin, Denny Jin, Shino Jomoto, Billie Jonn, Heewoo Jun, Tomer Kaftan, Lukasz Kaiser, Ali Kamali, Ingmar Kanitscheider, Nitish Shirish Keskar, Tabarak Khan, Logan Kilpatrick, Jong Wook Kim, Christina Kim, Yongjik Kim, Hendrik Kirchner, Jamie Ryan Kiros, Matthew Knight, Daniel Kokotajlo, Lukasz Kondraciuk, Andrew Kondrich, Aris Konstantinidis, Kyle Kosic, Gretchen Krueger, Vishal Kuo, Michael Lampe, Ikai Lan, Teddy Lee, Jan Leike, Jade Leung, Daniel Levy, Chak Ming Li, Rachel Lim, Molly Lin, Stephanie Lin, Mateusz Litwin, Theresa Lopez, Ryan Lowe, Patricia Lue, Anna Adeola Makanju, Kim Malfacini, Sam Manning, Todor Markov, Yaniv Markovski, Bianca Martin, Katie Mayer, Andrew Mayne, Bob McGrew, Scott Mayer McKinney, Christine McLeavey, Paul McMillan, Jake McNeil, David Medina, Aalok Mehta, Jacob Menick, Luke Metz, Andrey Mishchenko, Pamela Mishkin, Vinnie Monaco, Evan Morikawa, Daniel P. Mossing, Tong Mu, Mira Murati, Oleg Murk, David M'ely, Ashvin Nair, Reiichiro Nakano, Rajeev Nayak, Arvind Neelakantan, Richard Ngo, Hyeonwoo Noh, Ouyang Long, Cullen O'Keefe, Jakub W. Pachocki, Alex Paino, Joe Palermo, Ashley Pantuliano, Giambattista Parascandolo, Joel Parish, Emy Parparita, Alexandre Passos, Mikhail Pavlov, Andrew Peng, Adam Perelman, Filipe de Avila Belbute Peres, Michael Petrov, Henrique Pondé de Oliveira Pinto, Michael Pokorny, Michelle Pokrass, Vitchyr H. Pong, Tolly Powell, Alethea Power, Boris Power, Elizabeth Proehl, Raul Puri, Alec Radford, Jack Rae, Aditya Ramesh, Cameron Raymond, Francis Real, Kendra Rimbach, Carl Ross, Bob Rotsted, Henri Roussez, Nick Ryder, Mario D. Saltarelli, Ted Sanders, Shibani Santurkar, Girish Sastry, Heather Schmidt, David Schnurr, John Schulman, Daniel Selsam, Kyla Sheppard, Toki Sherbakov, Jessica Shieh, Sarah Shoker, Pranav Shyam, Szymon Sidor, Eric Sigler, Maddie Simens, Jordan Sitkin, Katarina Slama, Ian Sohl, Benjamin D. Sokolowsky, Yang Song, Natalie Staudacher, Felipe Petroski Such, Natalie Summers, Ilya Sutskever, Jie Tang, Nikolas A. Tezak, Madeleine Thompson, Phil Tillet, Amin Tootoonchian, Elizabeth Tseng, Preston Tuggle, Nick Turley, Jerry Tworek, Juan Felipe Cer'on Uribe, Andrea Vallone, Arun Vijayvergiya, Chelsea Voss, Carroll L. Wainwright, Justin Jay Wang, Alvin Wang, Ben Wang, Jonathan Ward, Jason Wei, CJ Weinmann, Akila Welihinda, Peter Welinder, Jiayi Weng, Lilian Weng, Matt Wiethoff, Dave Willner, Clemens Winter, Samuel Wolrich, Hannah Wong, Lauren Workman, Sherwin Wu, Jeff Wu, Michael Wu, Kai Xiao, Tao Xu, Sarah Yoo, Kevin Yu, Qiming Yuan, Wojciech Zaremba, Rowan Zellers, Chong Zhang, Marvin Zhang, Shengjia Zhao, Tianhao Zheng, Juntang Zhuang, William Zhuk, and Barret Zoph. 2023. Gpt-4 technical report.

Ayush Kumar Agrawal, Lester W. Mackey, and Adam Tauman Kalai. 2023. Do language models know when they're hallucinating references? *ArXiv*, abs/2305.18248.

Ebtesam Almazrouei, Hamza Alobeidli, Abdulaziz Alshamsi, Alessandro Cappelli, Ruxandra-Aimée Cojocaru, Daniel Hesslow, Julien Launay, Quentin Malartic, Daniele Mazzotta, Badreddine Noune, Baptiste Pannier, and Guilherme Penedo. 2023. The falcon series of open language models. *ArXiv*, abs/2311.16867.

Bar Lanyado. 2023. Can you trust chatgpt's package recommendations? `https://vulcan.io/blog/ai-hallucinations-package-risk/`.

Stella Biderman, Hailey Schoelkopf, Quentin G. Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. 2023. Pythia: A suite for analyzing large language models across training and scaling. *ArXiv*, abs/2304.01373.

Faeze Brahman, Sachin Kumar, Vidhisha Balachandran, Pradeep Dasigi, Valentina Pyatkin, Abhilasha Ravichander, Sarah Wiegreffe, Nouha Dziri, Khyathi Chandu, Jack Hessel, et al. 2024. The art of saying no: Contextual noncompliance in language models. *arXiv preprint arXiv:2407.12043*.

Ethan Chern, Steffi Chern, Shiqi Chen, Weizhe Yuan, Kehua Feng, Chunting Zhou, Junxian He, Graham Neubig, and Pengfei Liu. 2023. Factool: Factuality detection in generative ai - a tool augmented framework for multi-task and multi-domain scenarios. *ArXiv*, abs/2307.13528.

Thomas Claburn. 2024. Ai hallucinates software packages and devs download them – even if potentially poisoned with malware. *The Register*.

Ashwin Devaraj, William Sheffield, Byron Wallace, and Junyi Jessy Li. 2022a. Evaluating factuality in text simplification. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7331–7345, Dublin, Ireland. Association for Computational Linguistics.

Ashwin Devaraj, William Sheffield, Byron C. Wallace, and Junyi Jessy Li. 2022b. Evaluating factuality in text simplification. *Proceedings of the conference. Association for Computational Linguistics. Meeting*, 2022:7331–7345.

Esin Durmus, He He, and Mona Diab. 2020. FEQA: A question answering evaluation framework for faithfulness assessment in abstractive summarization. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 5055–5070, Online. Association for Computational Linguistics.

Nouha Dziri, Ehsan Kamalloo, Sivan Milton, Osmar Zaiane, Mo Yu, E. Ponti, and Siva Reddy. 2022. Faithdial: A faithful benchmark for information-seeking dialogue. *Transactions of the Association for Computational Linguistics*, 10:1473–1490.

Yanai Elazar, Akshita Bhagia, Ian H. Magnusson, Abhilasha Ravichander, Dustin Schwenk, Alane Suhr, Pete Walsh, Dirk Groeneveld, Luca Soldaini, Sameer Singh, Hanna Hajishirzi, Noah A. Smith, and Jesse Dodge. 2023. What's in my big data? *ArXiv*, abs/2310.20707.

Luyu Gao, Zhuyun Dai, Panupong Pasupat, Anthony Chen, Arun Tejasvi Chaganty, Yicheng Fan, Vincent Zhao, N. Lao, Hongrae Lee, Da-Cheng Juan, and Kelvin Guu. 2022. Rarr: Researching and revising what language models say, using language models. In *Annual Meeting of the Association for Computational Linguistics*.

Aaron Gokaslan and Vanya Cohen. 2019. Openwebtext corpus.

Dirk Groeneveld, Iz Beltagy, Pete Walsh, Akshita Bhagia, Rodney Kinney, Oyvind Tafjord, A. Jha, Hamish Ivison, Ian Magnusson, Yizhong Wang, Shane Arora, David Atkinson, Russell Authur, Khyathi Raghavi Chandu, Arman Cohan, Jennifer Dumas, Yanai Elazar, Yuling Gu, Jack Hessel, Tushar Khot, William Merrill, Jacob Daniel Morrison, Niklas Muennighoff, Aakanksha Naik, Crystal Nam, Matthew E. Peters, Valentina Pyatkin, Abhilasha Ravichander, Dustin Schwenk, Saurabh Shah, Will Smith, Emma Strubell, Nishant Subramani, Mitchell Wortsman, Pradeep Dasigi, Nathan Lambert, Kyle Richardson, Luke Zettlemoyer, Jesse Dodge, Kyle Lo, Luca Soldaini, Noah A. Smith, and Hanna Hajishirzi. 2024. Olmo: Accelerating the science of language models. *ArXiv*, abs/2402.00838.

Roger Baker Grosse, Juhan Bae, Cem Anil, Nelson Elhage, Alex Tamkin, Amirhossein Tajdini, Benoit Steiner, Dustin Li, Esin Durmus, Ethan Perez, Evan Hubinger, Kamil.e Lukovsiut.e, Karina Nguyen, Nicholas Joseph, Sam McCandlish, Jared Kaplan, and Sam Bowman. 2023. Studying large language model generalization with influence functions. *ArXiv*, abs/2308.03296.

Hangfeng He, Hongming Zhang, and Dan Roth. 2022. Rethinking with retrieval: Faithful large language model inference. *ArXiv*, abs/2301.00303.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Xiaodong Song, and Jacob Steinhardt. 2020. Measuring massive multitask language understanding. *ArXiv*, abs/2009.03300.

Karl Moritz Hermann, Tomas Kocisky, Edward Grefenstette, Lasse Espeholt, Will Kay, Mustafa Suleyman, and Phil Blunsom. 2015. Teaching machines to read and comprehend. *Advances in neural information processing systems*, 28.

Daniel Scott Himmelstein, Antoine Lizee, Christine Hessler, Leo Brueggeman, Sabrina L Chen, Dexter Hadley, Ari Green, Pouya Khankhanian, and Sergio E Baranzini. 2017. Systematic integration of biomedical knowledge prioritizes drugs for repurposing. *Elife*, 6:e26726.

Tamanna Hossain, Robert L. Logan IV, Arjuna Ugarte, Yoshitomo Matsubara, Sean Young, and Sameer Singh. 2020. COVIDLies: Detecting COVID-19 misinformation on social media. In *Proceedings of the 1st Workshop on NLP for COVID-19 (Part 2) at EMNLP 2020*, Online. Association for Computational Linguistics.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Delong Chen, Wenliang Dai, Andrea Madotto, and Pascale Fung. 2022. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55:1 – 38.